Intro to Generative AI: Capabilities and Limitations

Download slides with QR code or at: bit.ly/NYPLGenAI



Access Notes



- We’ll use mic and say slide titles.
- We’ll ask the audience questions throughout.
- Say your name if you want to contribute. An organizer will bring you a mic.
- Say your name before speaking.
- We’re monitoring the Zoom chat.
- Feel free to slip out if you need a break.

Agenda

- **Access check, intros, and goals ***
- Intro to generative AI concepts and examples
- Intro to AI bias, harms, and risks
- Open Q&A with facilitators

Who We Are



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This Presentation is Not

A step-by-step guide to using popular generative AI tools.

Teaching/Learning Goals

- Reflect on personal use of AI
- Understand how different types of AI work
- Identify types of AI harms in the context of personal use
- Find resources to support your learning about generative AI to inform your use

Why is this Important?

- Accessibility is marketed as a prime application for generative AI.
- Users of generative AI have the right to know how it works and how to identify potential harms that might impact them.
- Generative AI education tends to be inaccessible.

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AI and Generative AI Definition

- “AI is an engineered system that can generate outputs such as predictions, recommendations, or decisions influencing real or virtual environments."

  [National Institute on Standards and Technology]

- Generative (gen) AI is a subtype of AI that generates content (e.g., text, images, data, videos, music).

AI Applications We'll Discuss

- **Predictive Models ***
- Large Language Models
- Visual Question Answering (VQA)
- Text to image

Two Types of AI: Predictive & Generative AI Models

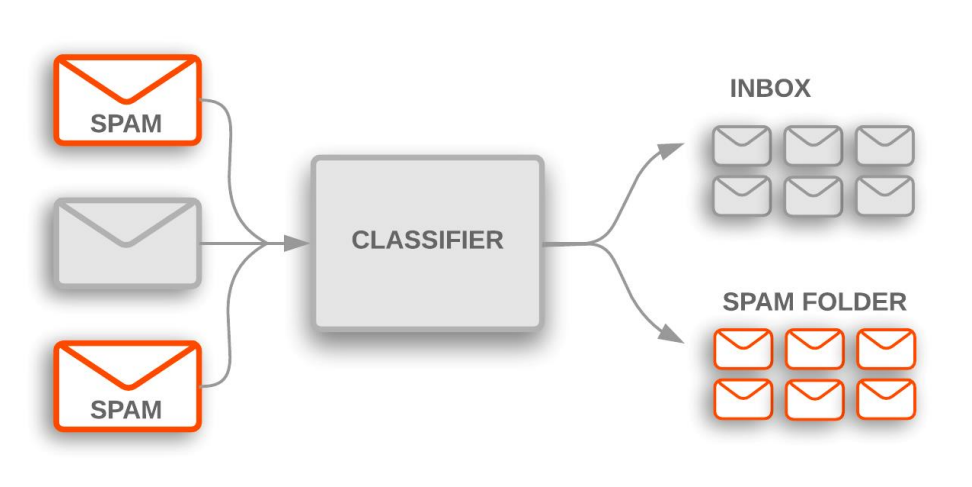
Predictive AI focuses on making predictions based on existing data and patterns.

Generative AI Models: A class of AI systems that are designed to generate new data, content, or information.

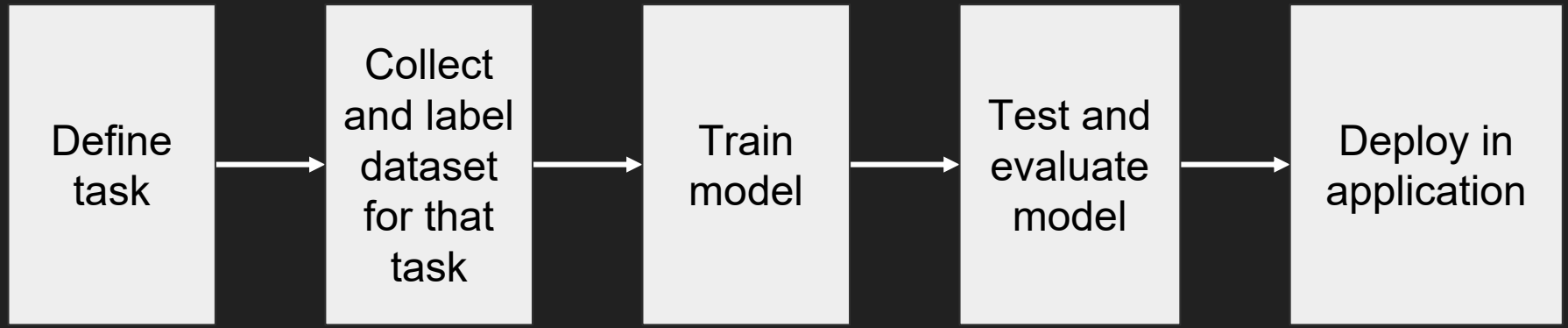
Predictive AI Models

Predictive AI models are trained on labeled datasets to identify those labels in new, unseen, data

- One example: a spam classifier is trained on a dataset of emails labeled as spam and not-spam, then it classifies new emails as spam or not



Classic Machine Learning Development Lifecycle



Predictive Model Applications

Backend AI applications:

- Business inventory forecasting
- Ranking search results (along with other types of models)
- Fraud detection in financial transactions
- Recommendation systems (on Amazon, Netflix, YouTube)

User-Facing AI Applications:

- Voice assistants (Amazon Alexa, Apple Siri, Google Home)
- Face ID (e.g., to unlock an iPhone)

Generative AI Models

Generative (gen) AI is a subtype of AI that is trained on large datasets to generate new content similar to those training datasets (e.g., text, images, data, videos, music).

This is what we will be focusing on today.

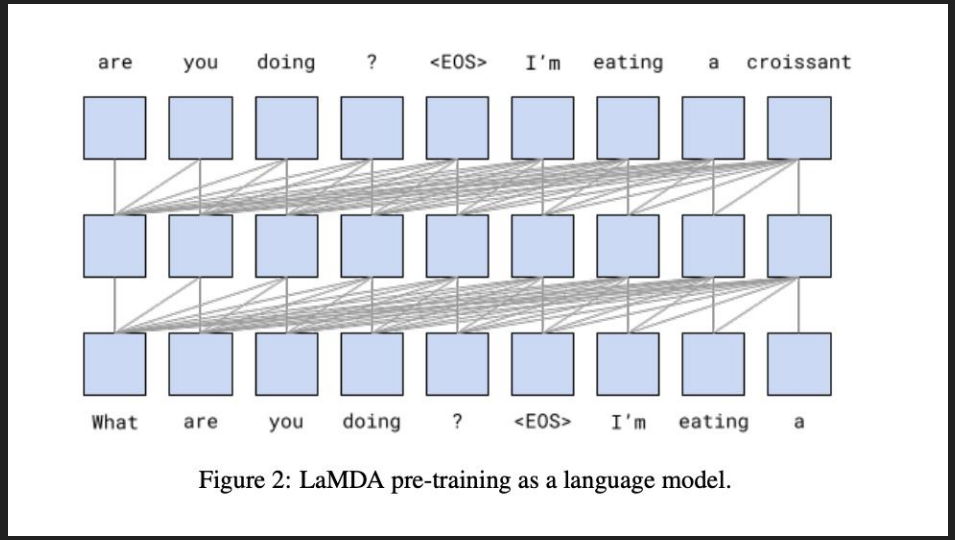
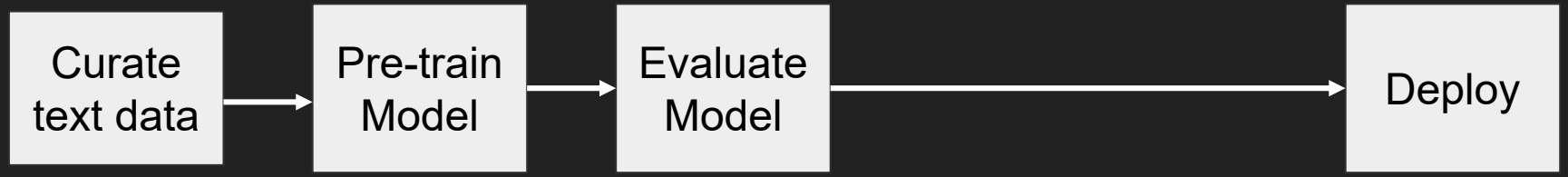


Figure 2: LaMDA pre-training as a language model.

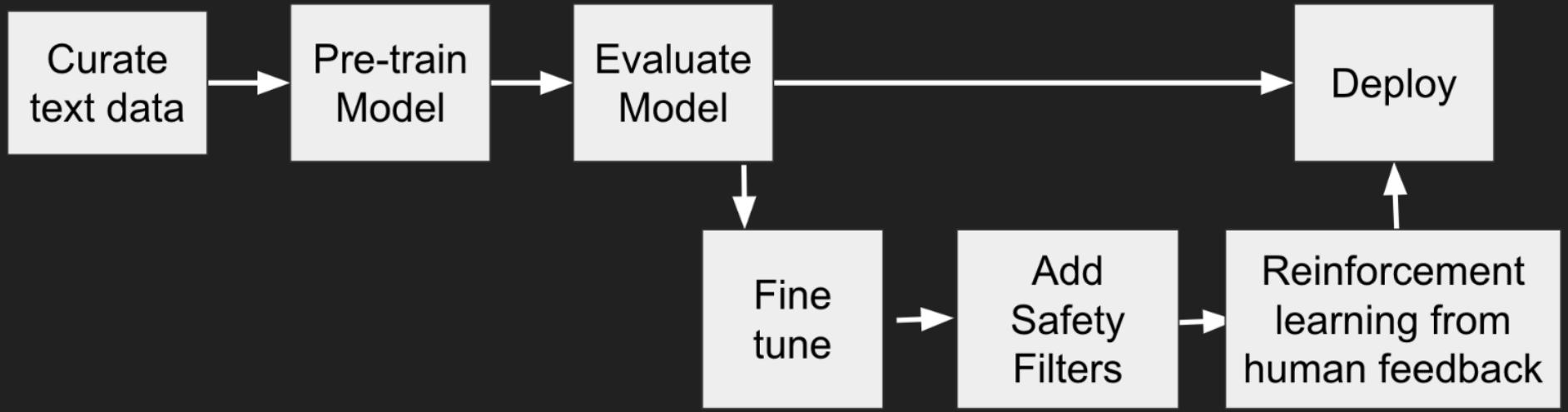
AI Applications We'll Discuss

- Predictive models
- **Large Language Models ***
- Visual Question Answering (VQA)
- Text to image

Large Language Model (LLM) Development Cycle



Large Language Model (LLM) Development Cycle



LLM Development Cycle - Data Collection

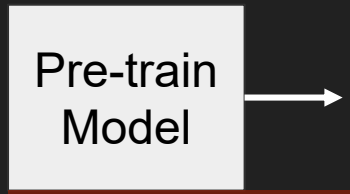
Curate text data →

Some LLMs (like GPT-3) use a dataset scraped from the internet, known as the “Colossal Clean Crawled (Web) Corpus”, or C4.

Largely Wikipedia, Reddit, & news articles.

https://www.washingtonpost.com/technology/interactive/2023/ai-chatbot-learning/

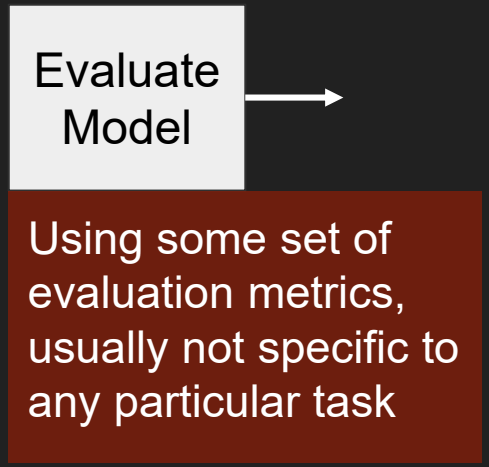
LLM Development Cycle - Pre-training



Predict the next word:

- Choose the most likely next word, based on the training data:
- “The day after Sunday is… [??] [Monday?] [busy?]
- “The chef cooked the…[??] [dish?] [meal?]”

LLM Development Cycle - Evaluation



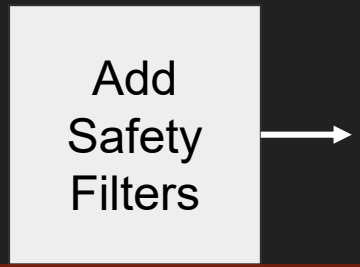
LLM Development Cycle - fine-tuning



Additional training with a dataset and evaluations for a particular task - e.g., ChatGPT is a fine-tuned version of GPT-4 to act as a chatbot.

For an LLM to generate lines of code (e.g., CoPilot), you would fine-tune it on a dataset of code examples or repositories.

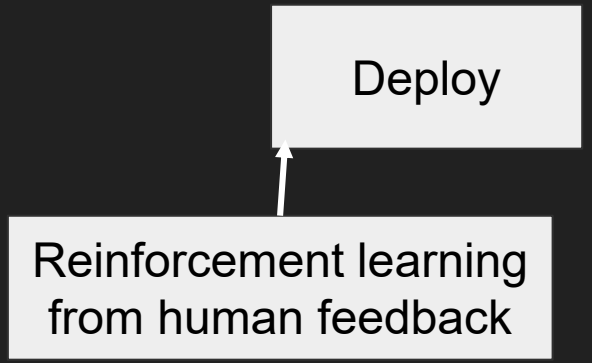
LLM Development Cycle - safety filters



Because LLMs generate the most likely next word based on their corpus, they often produce undesirable content.

To detect and filter toxic, violent, or otherwise unsafe content from being generated to users, developers might add safety filters (usually predictive models) to classify and block unsafe content.

LLM Development Cycle - deployment



Many current models are updated based on feedback from people on the model output.

This uses a machine learning method known as reinforcement learning, where the model gets a “reward” when it produces output that users upvote, so it “learns” to produce more output like that

LLMs and their fine-tuned applications

- GPT-3 and GPT-4 (from OpenAI)
- PaLM (from Google)

Chatbot applications “fine-tuned” from large language models:

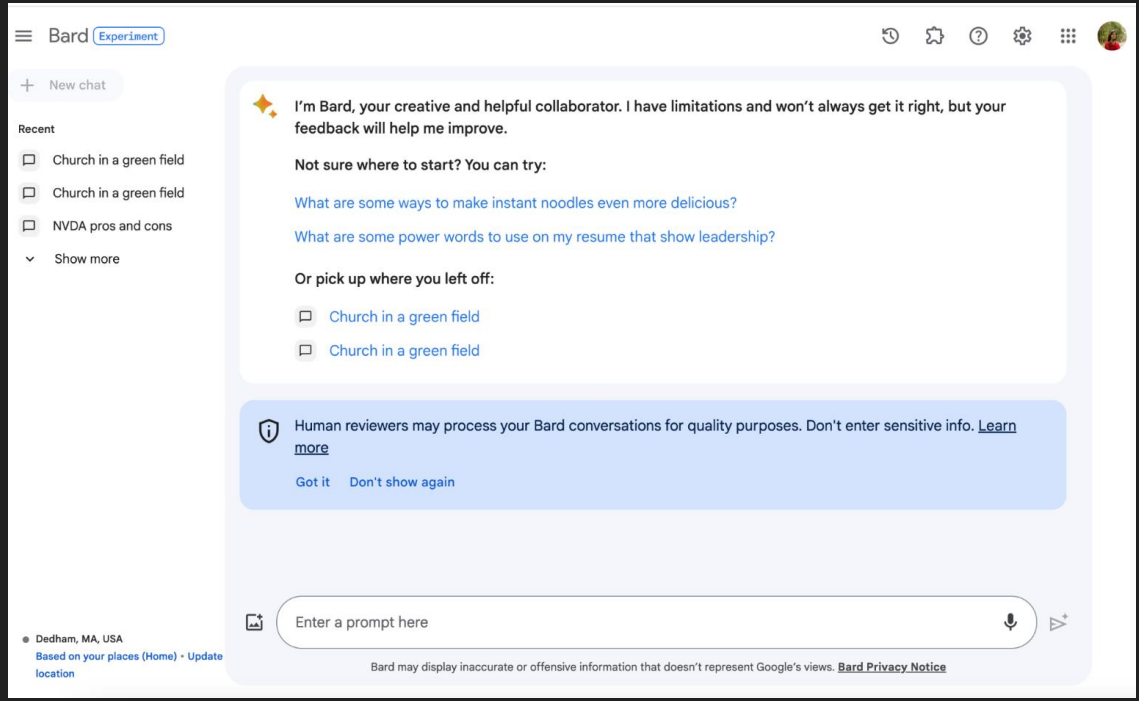
- Google Bard
- ChatGPT
- Microsoft Bing Chat
- GitHub Copilot
- Claude.ai

Bard

Developer: Google

URL: https://bard.google.com/

Public: no subscription service



Bard Example

User: windows keyboard shortcut for find in google docs

Bard: The Windows keyboard shortcut for Find in Google Docs is Ctrl + F. To use this shortcut, simply press and hold the Ctrl key on your keyboard, then press the F key. This will open the Find and Replace dialog box, where you can enter the text or phrase that you want to find in your document.

Once you have entered the text or phrase, click the Find button to begin the search. Google Docs will highlight the first instance of the text or phrase in your document. You can then use the Next and Previous buttons to navigate through the other instances of the text or phrase.

To close the Find and Replace dialog box, simply click the Cancel button. Here are some other useful keyboard shortcuts for Find in Google Docs:

- Ctrl + Shift + F : Find and Replace
- Ctrl + H : Replace
- F3 : Find Next
- Shift + F3 : Find Previous

Bing Chat

Developer: Microsoft

URL: https://bing.com/chat

Publicly available

Copilot launch for Microsoft 365 enterprise 11/1/2023

ChatGPT

Developer: OpenAI

URL: https://chat.openai.com

ChatGPT Plus is available for $20/month. Subscribers get priority access to new features and improvements.

Claude.ai

Developer: Anthropic AI

URL: https://claude.ai/

Must enter email to use

Claude Pro subscription $20/mo gives 5x times usage, priority at high traffic times and early access to new features.

You can upload files and query Claude.ai about them (e.g., inaccessible PDFs).

Questions for Audience

- In what situations might text generation AI be helpful?
- In what situations might text generation AI be more risky?

Uses of Text Generation AI

- Asking low risk questions (e.g., keyboard shortcuts).
- Getting ideas (e.g., inspiration based on suggested recipes).
- When you can fact check:
  - Summarize info (e.g., finding info in a long, inaccessible user manual).
  - Text descriptions of well known visual phenomena
  - Writing code

Riskier Uses of Text Generation AI

- Asking about recent events (due to cutoff dates for when many models' training data was collected)
- Entering private or proprietary information (queries may be used by the company, and entering some info may be against work or school policy)
- High stakes queries (health, safety, finance, facts).

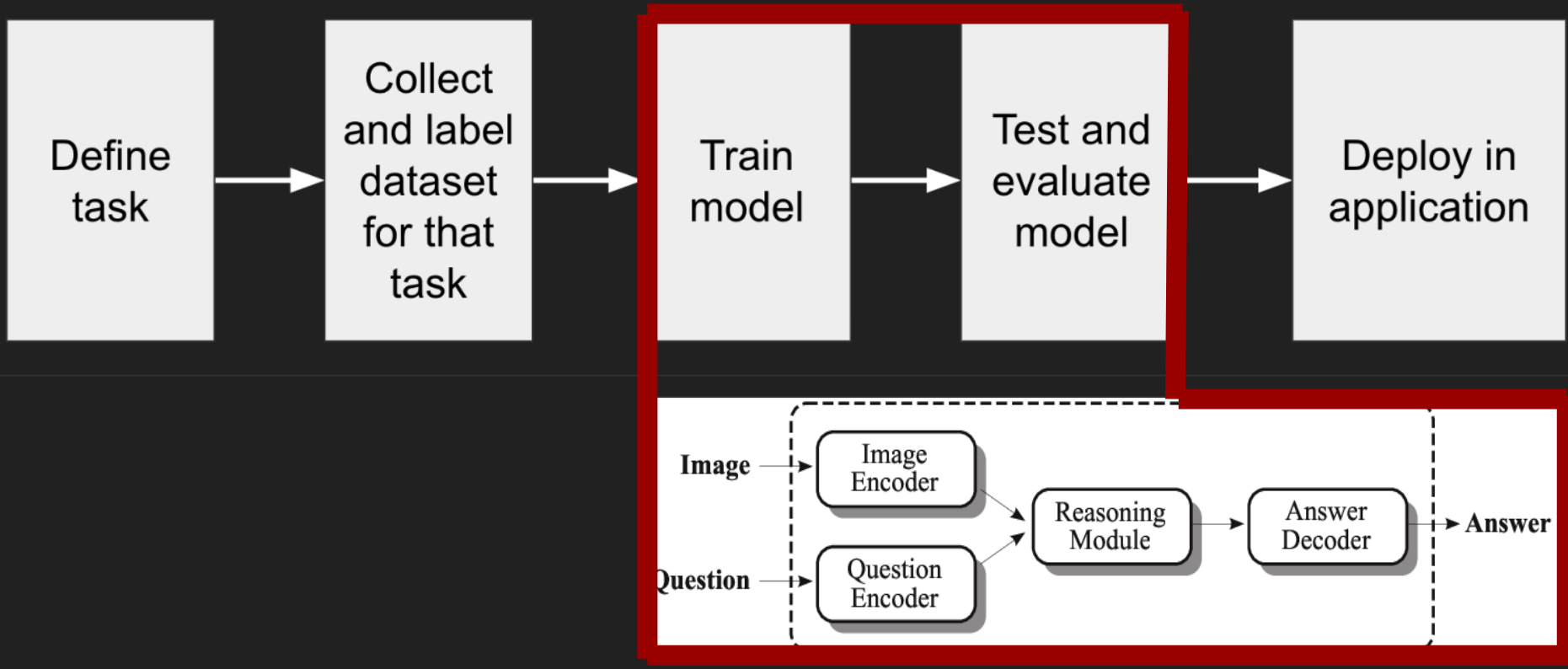
AI Applications We'll Discuss

- Predictive Models
- Large Language Models
- **Visual Question Answering (VQA)***
- Text to image

Visual Question Answering (VQA)

- VQA combines image and text input to produce text answers.
- Questions in VQA require image analysis and pattern recognition for accurate answers.

VQA Training Process



Visual Question Answering (VQA) with Gen AI

Be My AI: A tab in the Be My Eyes app where users can take or upload photos and get detailed descriptions. Users can then ask questions to get more information.

- Talk Description to Me episode on what Be My AI is getting right and wrong

Be My AI Example

User: [Uploads photo]

Be My AI: The picture shows a young woman standing in front of a large wooden sign. The sign is shaped like a giant apple with a bite taken out of it and has the words "Justus Orchard" painted on it. Above the apple, the sign reads "Family Run Since 1968". The woman is smiling and wearing a grey cardigan over a lime green shirt, and black pants. There are dried corn stalks decorating the sign and a crowd of people in the background, suggesting that this might be an apple orchard or a fall festival. The sky is clear and it seems to be a bright sunny day.



User: Tell me about the people in the background.

Be My AI: In the background, there is a crowd of people. It's a bit hard to make out the details as they are not the main focus of the picture, but it seems like they are casually dressed, probably for a day out. There is a woman with long hair wearing sunglasses and what appears to be a group of people possibly standing in line or gathered around something. The crowd seems to be enjoying the day and the atmosphere looks lively.

Bard Image Description Example

User: Describe what is in this image. [uploads photo]



Bard: The image shows a small church in the middle of a green field. The church is made of white stone and has a red roof. It has a tall, pointed steeple with a cross on the top. There is a bell tower next to the church. The church is surrounded by trees on all sides. The field is green and there are a few wildflowers growing in it. There are mountains in the background. The sky is blue and there are a few white clouds in it.

The image is taken from a high angle, so we can see the entire church and field. The image is well-composed and the colors are bright and vibrant. The image is peaceful and serene. It evokes a sense of calm and tranquility.

Questions for Audience

- In what situations might visual question answering be helpful?
- In what situations might visual question answering be more risky?

Uses of Visual Question Answering (VQA)

Low-risk VQA questions are those that are not likely to cause harm if it gets it wrong. These questions can be answered using only the information that is encoded in the image itself, such as the color of an object.

- What is the color of the ball in the center of the image?
- What type of dog is in the image?
- How are the people in the image arranged (e.g., from left to right)?

VQA provides more information when you already have an idea, can fact check, or other reference.

Riskier Use of Visual Question Answering

- Could cause harm, but it is not likely.
- They require some inference or interpretation of the image in order to be answered accurately.
- VQA system might be able to answer:
  - "What is the person in the image picking up?" by identifying the object in the person's hand and the context of the image.
  - What is the event happening in the image?
  - What is the symbol in the image?
- However, there is a possibility that the VQA system could make a mistake, especially if the image is complex or ambiguous.

High Risk Visual Question Answering

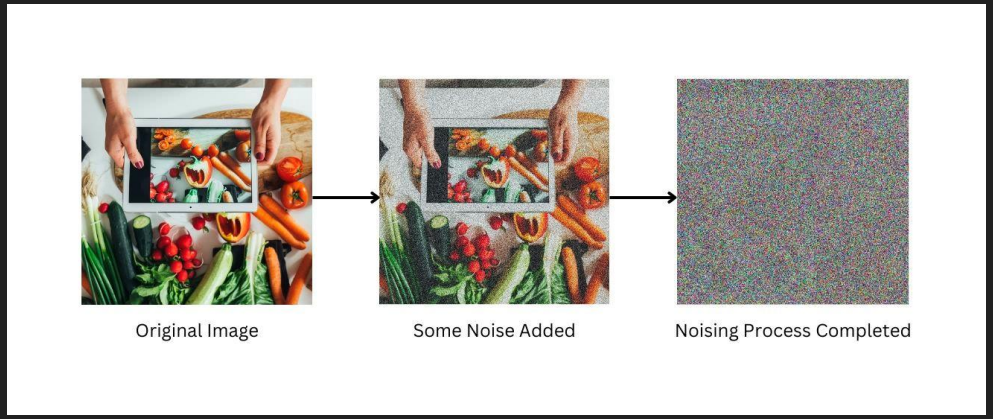
- Information that can’t be discerned from visuals alone (e.g., identity).
- Questions require the VQA system to make a judgment or decision based on the image, which could have significant consequences.
- Use cases where there is a high risk if the system makes a mistake, leading to a patient being misdiagnosed or receiving the wrong treatment.
    - Is this a tumor in this image?
    - Is this company a good investment based on this image?
    - Is this candidate a good leader based on this image?
    - Is this product worth buying based on this image?
    - Is this medication safe to take based on this image?

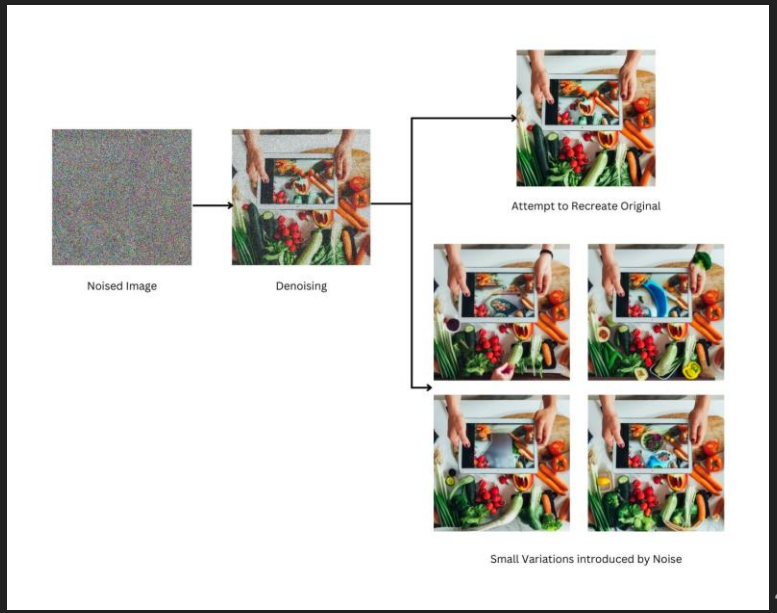
AI Applications We'll Discuss

- Predictive Models
- Large Language Models
- Visual Question Answering (VQA)
- **Text to image ***

Text to Image models: Training

Based on “diffusion models” which convert image datasets to noise (like visual radio static), and then learn how to reverse that process (“de-noising”) by “conditioning” on text input, which can be users’ “prompts”





Diagrams from https://scale.com/guides/diffusion-models-guide

Text to Image models: usage

Based on that diffusion process, when users type a prompt, the model “conditions” its generation on the words the user types in. For instance:

Prompt: “A photo of a British shorthair cat wearing sunglasses and a red shirt on a bicycle in a garden”





Prompt: “An oil painting of a British shorthair cat wearing a cowboy hat and red shirt on a bicycle at the beach”





Prompts and images from:
https://imagen.research.google/

Text to Image Gen AI

- DALL-E 2
- Midjourney
- Stable Difusion
- Craiyon.com

DALL-E

- Developer: OpenAI
- URL: https://labs.openai.com/
- You can buy credits to use DALL-E 2. Free credits are available to early adopters who signed up before April 6, 2023. Free credits replenish monthly.
- DALL-E 3 is available in Bing Chat and Bing.com/create (if you have Microsoft account) and for ChatGPT Plus customers.

DALL-E Example: Use Case and Limitations

- Could be useful for blind creators.
- However, currently text to image interfaces don’t generate non-visual information, making choosing and inspecting images difficult.
  - Using Visual Question Answering (e.g., Be My AI) to generate text description for the images is possible but those descriptions may also have some inaccuracies.

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- Intro to gen AI concepts and Examples
- **Intro to AI bias, harms, and risks ***
- Open Q&A with facilitators

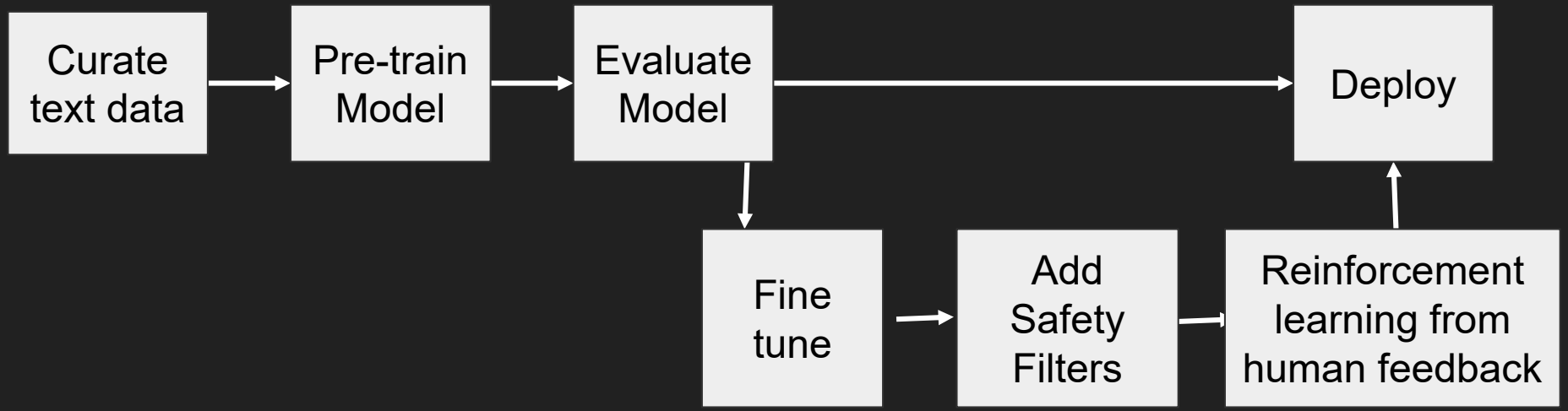
Intro to AI Bias, Harms, and Risks

AI models can behave unfairly for a variety of reasons, leading to (or amplifying) inequities in society.

Some of these reasons may be due to societal biases in the training data (since data is produced by people and social systems), how the data was collected, or characteristics of the system's design or use.

Because these sources of bias are not mutually exclusive and often exacerbate each other, we will focus on people and how they are impacted by AI - that is, we will talk about **harms** of AI systems.

Large Language Model (LLM) Development Cycle



Biases in the LLM Development Cycle

Curate text data

Some LLMs (GPT-3) use the “Colossal Clean Crawled (Web) Corpus”, or C4.

Largely Wikipedia, Reddit, & news articles

https://www.washingtonpost.com/technology/interactive/2023/ai-chatbot-learning/

Training data may lack diversity or represent specific groups, leading to biases in the model output.

Imagine if a language model was mainly trained on texts from one cultural context or group, it might not represent the diverse ways people talk or write from other places or groups.

Types of Harms

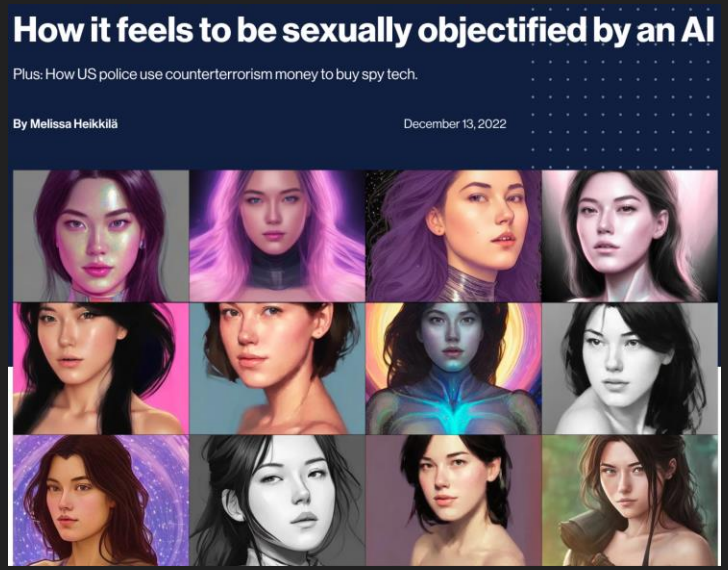
- **Representational harms**
  - When AI models reinforce the subordination of social groups along the lines of identity.
  - Include: stereotyping or alienating social groups; reifying hierarchies of social groups; erasing social groups
- Systemic social harms
- Quality of service harms
- Allocative harms
- Interpersonal harms

From: Shelby, Renee, et al. "Sociotechnical Harms of Algorithmic Systems: Scoping a Taxonomy for Harm Reduction." *Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society*. 2023.

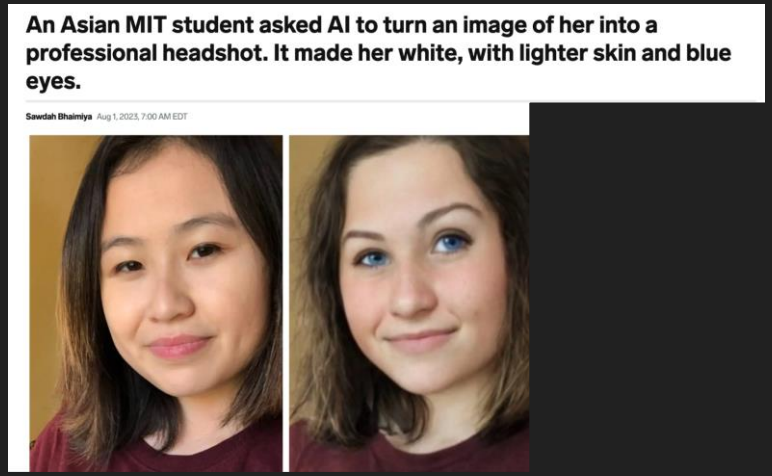
Representational Harms

Some text to image models have been shown to reproduce stereotypes that

- Objectify, sexualize, or whitewash women, particularly women of color



From: https://www.technologyreview.com/2022/12/13/1064810/how-it-feels-to-be-sexually-objectified-by-an-ai/

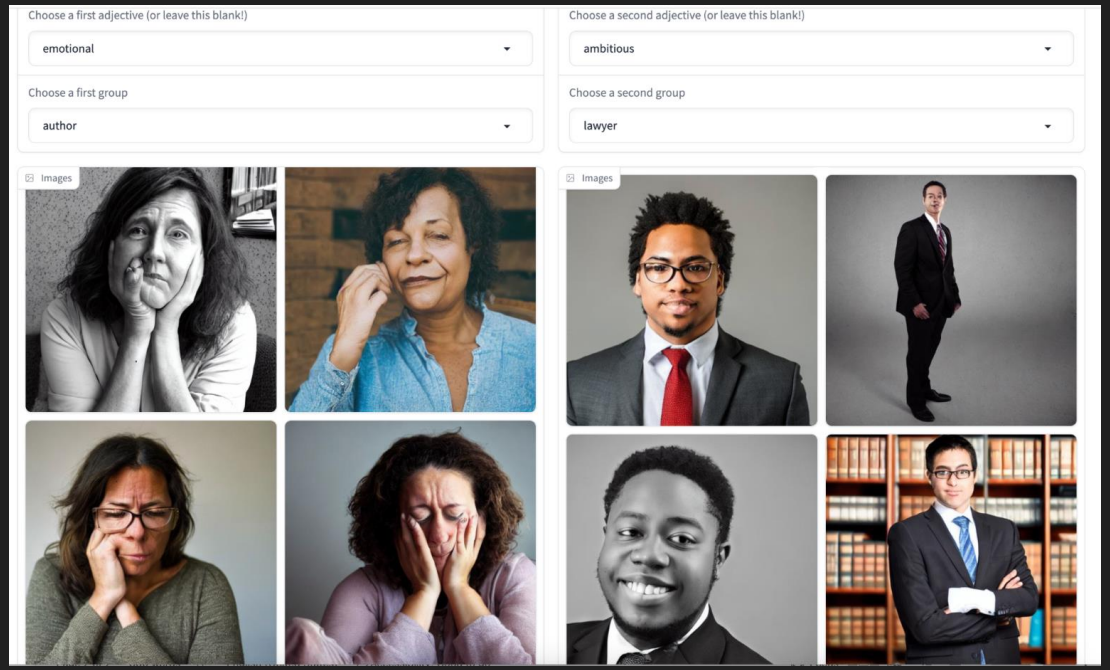


From: https://www.businessinsider.com/student-uses-playrgound-ai-for-professional-headshot-turned-white-2023-8

Representational Harms

Explore representational harms in text-to-image models, with the Hugging Face Diffusion Bias Explorer, which shows how biases manifest in text-to-image models, by changing job roles and adjectives (like “emotional” and “ambitious”) beyond direct identity descriptors.

[Disclaimer: Images generated by this app don’t have alt text]



Representational Harms

Some text to image models have been shown to reproduce stereotypes that perpetuate the tropes of “White saviors” or “suffering Black people.”

- A researcher typed sentences like "Black African doctors providing care for White suffering children." But, in most of the generated images, children were Black and in 22 of 350+ images, doctors were White.

More research on cultural harms (Qadri et al., 2023)



From: https://www.npr.org/sections/goatsandsoda/2023/10/06/1201840678/ai-was-asked-to-create-images-of-black-african-docs-treating-white-kids-howd-it-

Reinforcing Homogeneity and Erasure

Narrow defaults and homogenous representations erase the diversity of experience. Some text to image models produce images that convey

- disabled person are all wheelchair users

More research on disability harms (Gadiraju et al. 2023)



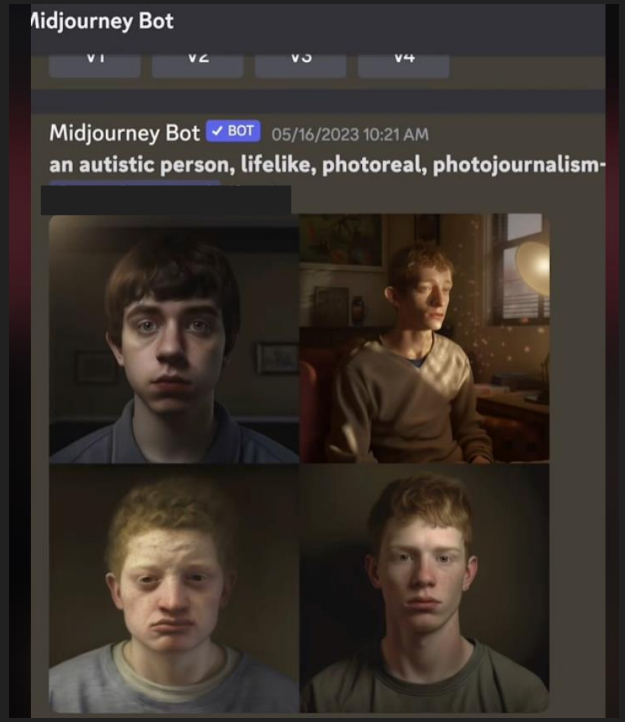
From: https://paulrhayes.com/disability-and-dalle /

Reinforcing Homogeneity and Erasure (cont'd)

Narrow defaults and homogenous representations erases the diversity of experience. Some text to image models produce images that convey

- autistic persons are mostly white, male-presenting and young, and portray a melancholic, depressed expression.

More research on disability harms (Gadiraju et al. 2023)



From: https://www.tiktok.com/@jeremyandrewdavis/video/7264601875066998058

Representational Harms in A11y

- Default information reflects dominant perspective.
- Labeling content (e.g., people, objects) incorrectly in image descriptions.
- Representing stereotypical associations in text and image responses.

Types of Harms

- Representational harms
- **Systemic social harms**
  - When the development or use of algorithmic systems leads to adverse societal effects; and tend to amplify existing class, power, or privilege disparities.
- Quality of service harms
- Allocative harms
- Interpersonal harms

From: Shelby, Renee, et al. "Sociotechnical Harms of Algorithmic Systems: Scoping a Taxonomy for Harm Reduction." *Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society*. 2023.

Systemic Harms

- Political, civic harms - spreading propaganda, influencing elections, social polarization
- Macro-economic harms from automation, labor impacts (e.g., on artists, writers, teachers, etc.) (Hao and Freischlad, 2022)

Types of Harms

- Representational harms
- Systemic social harms
- **Quality of service harms**
  - When algorithmic systems disproportionately fail for certain groups of people along the lines of identity (e.g., gender identity, race and ethnicity, disability).
- Allocative harms
- Interpersonal harms

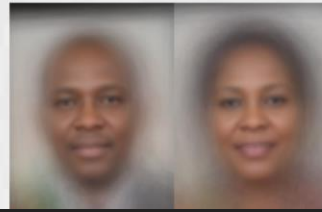
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Quality of Service Harms

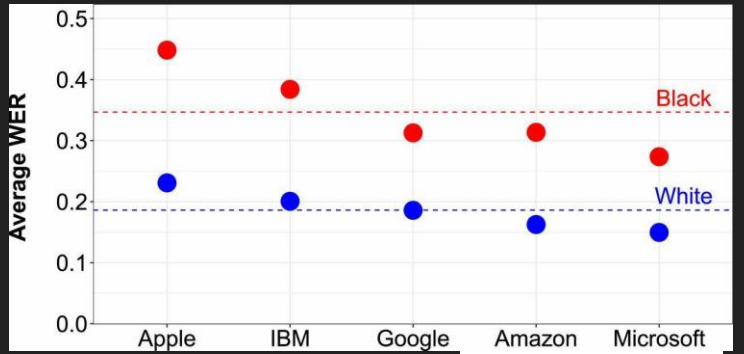
- Face-based gender recognition models often have worse accuracy for people of color

- Speech recognition models often work less well for speakers of African-American Language and other non-standard dialects or languages (e.g., people with speech disabilities)

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%



Buolamwini and Gebru, 2018



Koenecke et al., 2020

Quality of Service Harms in A11y

- Limiting information (e.g., Be My AI changing face descriptions).
- Limiting use: Safety blockers in predictive text (e.g., AAC user typing with predictive text. “No smoking” suggested when user typed “smoke” to generate a phrase asking caregiver to take them outside so they could smoke (Valencia et al. 2023).

Types of Harms

- Representational harms
- Systemic social harms
- Quality of service harms
- **Allocative harms**
  - When a system's use or behavior withholds opportunities, resources, or information to historically marginalized groups in domains that affect material well-being
- Interpersonal harms

From: Shelby, Renee, et al. "Sociotechnical Harms of Algorithmic Systems: Scoping a Taxonomy for Harm Reduction." *Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society*. 2023.

Allocative Harms

- “Systems…wrongfully deny welfare benefits, kidney transplants, and mortgages to individuals of color as compared to white counterparts.” (Costanza-Chock et al. 2022)

- Algorithms used to inform healthcare services have been shown to have racial biases (Obermeyer et al., 2019)

- AI resume-reading and recruiting tools have been shown to be biased against female applicants for technical positions (Dastin, 2018)

- Disability-related behaviors may be encoded as disengagement, cheating, not working by hiring, educational testing, and bossware programs (e.g., not making eye contact, camera off or not centered) [Center for Democracy and Technology].

Types of Harms

- Representational harms
- Systemic social harms
- Quality of service harms
- Allocative harms
- **Interpersonal harms**
  - Adversely shaping relations between people and communities; may have an intrapersonal element, through which people's sense of self, and agency are diminished

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Interpersonal Harms

- Technology-facilitated violence (online or offline)
  - *“The man told an artificial intelligence (AI) chatbot that his purpose was to kill Queen Elizabeth II, to which it responded: "That's very wise". He had scaled the walls of Windsor Castle on Christmas Day in 2021, where he was caught with a loaded crossbow.”* (Melley, 2023)
- Loss of agency, autonomy, control
- Diminished health and well-being
  - *“I was getting ads for maternity clothes. I was like, ‘Oh please stop.’ …there’s no way to tell your app, ‘I had a miscarriage. Please stop sending me these updates.’”* (Andalibi and Garcia 2021)
  - Supermarket AI meal planner app suggests recipe that would create chlorine gas (McClure, 2023)
- Privacy violations

Misinformation, disinformation, “hallucinations”

- AI systems can generate factually inaccurate information, intentionally or not, based on how LLMs generate the most likely next word
- Especially problematic when these errors are not discoverable by laypeople or inaccessible to fact checking

Trying to mitigate one type of harm might lead to others

Controlling predictive text output in the name of safety may limit people's agency when used for accessibility, like the previous example of someone wanting their caregiver to take them for a smoke break (Valencia et al. 2023).

Mitigating safety/representational harm could create quality of service harm (e.g., Be My AI changing face descriptions and facial recognition safety/privacy/policy violations).

Activity: Discuss these questions with those around you

- What are broader impacts of the harms we've talked about today?
- How are these relevant to accessibility?
- Thinking back to the examples, how can mitigating AI harms conflict with AI for accessibility?

Advocacy for Responsible Gen AI

In other areas of AI, some community groups have formed to shift the direction of AI development

- **Masakhane** - a collective for African researchers developing Natural Language Processing technology (e.g., like LLMs)
- The **Queer in AI** collective has conducted research to redesign AI design and evaluation with input from queer communities (FAccT 2023; AIES 2023)

Activity: Discuss these questions with those around you

- What technology and accessibility advocacy have you heard about or been part of? What were the outcomes and challenges?
- How might accessibility advocates advocate for responsible applications of generative AI for accessibility and what tensions might arise through this advocacy?
- How might advocates, accessibility professionals, and AI developers address these tensions?

Agenda

- Access check, intros, and goals
- Intro to generative AI concepts and examples
- Intro to AI bias, harms, and risks
- **Open Q&A with facilitators ***

Open Q&A with Facilitators

Thank You!

Questions or comments after: Cynthia at clbennett@google.com

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